

341. In the metallurgy of zinc, the zinc dust obtained from roasting and reduction of zinc sulphide contains some ZnO. How is this removed

- (a) Absorbance of ultraviolet light and re-emission of white light is employed
- (b) Shock cooling by contact with a shower of molten lead is done
- (c) X-ray method is used
- (d) Smelting is employed

Answer: D — (d) Smelting is employed

Chemical reason: The correct choice is Smelting is employed. It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.

342. In the metallurgy of copper, metallic copper is finally formed in the furnace by the reactions

- (a) $\text{Cu}_2\text{S} + \text{O}_2 \rightarrow 2\text{Cu} + \text{SO}_2$
- (b) $2\text{CuS} + 3\text{O}_2 \rightarrow 2\text{CuO} + 2\text{SO}_2$ $2\text{CuO} + \text{CuS} \rightarrow 3\text{Cu} + \text{SO}_2$
- (c) $2\text{Cu}_2\text{S} + 3\text{O}_2 \rightarrow 2\text{Cu}_2\text{O} + 2\text{SO}_2$ $\text{Cu}_2\text{S} + 2\text{Cu}_2\text{O} \rightarrow 6\text{Cu} + \text{SO}_2$
- (d) $\text{CuS} + \text{O}_2 \rightarrow \text{Cu} + \text{SO}_2$

**Answer: C — (c) $2\text{Cu}_2\text{S} + 3\text{O}_2 \rightarrow 2\text{Cu}_2\text{O} + 2\text{SO}_2$
 $\text{Cu}_2\text{S} + 2\text{Cu}_2\text{O} \rightarrow 6\text{Cu} + \text{SO}_2$**

Chemical reason: Applying the known reaction of the stated transition-metal compound under these reagents/conditions gives $2\text{Cu}_2\text{S} + 3\text{O}_2 \rightarrow 2\text{Cu}_2\text{O} + 2\text{SO}_2$ $\text{Cu}_2\text{S} + 2\text{Cu}_2\text{O} \rightarrow 6\text{Cu} + \text{SO}_2$. The

choice is consistent with the expected oxidation state and stability of the product.

343. In the smelting of roasted copper pyrites ore, melting occurs so that the first reaction is

- (a) All the sulphur preferentially combines with iron to form FeS and CuO is formed
- (b) All the sulphur preferentially combines with copper to form CuS and FeO is formed
- (c) All the sulphur preferentially combines with iron to form FeS and Cu_2O is formed
- (d) All the sulphur preferentially combines with copper to form Cu_2S and FeO is formed

Answer: A — (a) All the sulphur preferentially combines with iron to form FeS and CuO is formed

Chemical reason: This is a standard nomenclature/composition identification: the name or formula given in the question corresponds to All the sulphur preferentially combines with iron to form FeS and CuO is formed. The choice is fixed by the known composition of that compound/mineral.

344. In the oxidation of Cu, the reaction which takes place in bessemer converter is

- (a) $2\text{CuFeS}_2 + \text{O}_2 \rightarrow \text{Cu}_2\text{S} + \text{FeS} + \text{SO}_2$
- (b) $2\text{Cu}_2\text{S} + 3\text{O}_2 \rightarrow 2\text{Cu}_2\text{O} + 2\text{SO}_2$
- (c) $2\text{Cu}_2\text{O} + \text{Cu}_2\text{S} \rightarrow 6\text{Cu} + \text{SO}_2$
- (d) $2\text{FeS} + 3\text{O}_2 \rightarrow 2\text{FeO} + 2\text{SO}_2$

Answer: C — (c) $2\text{Cu}_2\text{O} + \text{Cu}_2\text{S} \rightarrow 6\text{Cu} + \text{SO}_2$

Chemical reason: Applying the known reaction of the stated transition-metal compound under these reagents/conditions gives $2\text{Cu}_2\text{O} + \text{Cu}_2\text{S} \rightarrow$



$6\text{Cu} + \text{SO}_2$. The choice is consistent with the expected oxidation state and stability of the product.

345. Silica is added to roasted copper ore during smelting in order to remove

- (a) Cuprous sulphide
- (b) Cuprous oxide
- (c) Ferrous oxide
- (d) Ferrous sulphide

Answer: C — (c) Ferrous oxide

Chemical reason: This is a standard nomenclature/composition identification: the name or formula given in the question corresponds to Ferrous oxide. The choice is fixed by the known composition of that compound/mineral.

346. Parke's process is used to extract

- (a) Silver using NaCN
- (b) Copper using CuFeS_2
- (c) Silver from argentiferous lead
- (d) Silver by forming amalgam

Answer: C — (c) Silver from argentiferous lead

Chemical reason: Parkes' process extracts silver from argentiferous lead by exploiting the greater solubility of silver in molten zinc than in molten lead.

347. Identify the reaction that doesn't take place during the smelting process of copper extraction

- (a) $2\text{FeS} + 3\text{O}_2 \rightarrow 2\text{FeO} + 2\text{SO}_2 \uparrow$
- (b) $\text{Cu}_2\text{O} + \text{FeS} \rightarrow \text{Cu}_2\text{S} + \text{FeO}$
- (c) $2\text{Cu}_2\text{S} + 3\text{O}_2 \rightarrow 2\text{Cu}_2\text{O} + 2\text{SO}_2 \uparrow$
- (d) $\text{FeO} + \text{SiO}_2 \rightarrow \text{FeSiO}_3$

Answer: C — (c) $2\text{Cu}_2\text{S} + 3\text{O}_2 \rightarrow 2\text{Cu}_2\text{O} + 2\text{SO}_2 \uparrow$

Chemical reason: Applying the known reaction of the stated transition-metal compound under these reagents/conditions gives $2\text{Cu}_2\text{S} + 3\text{O}_2 \rightarrow 2\text{Cu}_2\text{O} + 2\text{SO}_2 \uparrow$. The choice is consistent with the expected oxidation state and stability of the product.

348. The extraction of which of the following metals involves bessemerisation

- (a) Fe
- (b) Ag
- (c) Al
- (d) Cu

Answer: D — (d) Cu

Chemical reason: Bessemerisation is used in copper metallurgy to oxidize iron sulfide and part of copper sulfide in copper matte, ultimately producing blister copper.



349. Bessemer converter is used for

- (a) Steel
- (b) Wrought iron
- (c) Pig iron
- (d) Cast iron

Answer: C — (c) Pig iron

Chemical reason: The Bessemer converter is classically used in steel manufacture to oxidize carbon and impurities from molten pig iron.

350. In the cyanide process for the extraction of silver, sodium cyanide is used to

- (a) Convert silver into a soluble silver complex
- (b) Reduce silver
- (c) Precipitate silver
- (d) Oxidise silver

Answer: A — (a) Convert silver into a soluble silver complex

Chemical reason: In cyanide extraction, CN^- converts silver into the soluble complex $[\text{Ag}(\text{CN})_2]^-$. This transfers Ag from the ore into solution so it can later be recovered.

351. Parke's process of desilverization of lead depends upon

- (a) Partition coefficient of silver between molten zinc/molten lead having a high value
- (b) Partition coefficient of silver between molten zinc/molten lead having a low value
- (c) Crystallizing out of pure lead while the silver-lead eutectic which has a lower melting point is left behind in liquid form
- (d) Chemical combination of zinc and silver which precipitates out easily

Answer: A — (a) Partition coefficient of silver between molten zinc/molten lead having a high value

Chemical reason: Parkes' process depends on silver having a strong preference for molten zinc over molten lead. The effective partition coefficient into the zinc phase is therefore high.

352. Which is impure form of iron

- (a) Cast iron
- (b) Wrought iron
- (c) Steel iron
- (d) None

Answer: A — (a) Cast iron

Chemical reason: Cast iron is an impure commercial form of iron with a high carbon content and other impurities. Wrought iron is much purer.



353. Blister copper is

- (a) Pure copper
- (b) Ore of copper
- (c) Alloy of copper
- (d) 1% impure copper

Answer: D — (d) 1% impure copper

Chemical reason: Blister copper is about 98–99% Cu and contains roughly 1% impurities. Its blistered surface results from gas evolution during solidification.

354. In the extraction of copper when molten copper is cooled slowly, blister copper is obtained due to evolution of the following gas

- (a) Water vapour
- (b) Sulphur dioxide
- (c) Carbon dioxide
- (d) Carbon monoxide

Answer: B — (b) Sulphur dioxide

Chemical reason: During conversion, SO_2 dissolves in molten copper and escapes as the metal cools, producing blisters on the surface; hence the name “blister copper.”

355. In the electrolytic purification of copper some gold is found in the

- (a) Cathode
- (b) Cathode mud
- (c) Anode mud
- (d) Electrolyte

Answer: C — (c) Anode mud

Chemical reason: Noble metal impurities such as gold do not dissolve at the copper anode

during electrolytic refining. They settle below the anode as anode mud.

356. Purest form of iron is

- (a) Cast iron
- (b) Wrought iron
- (c) Hot steel
- (d) Stainless steel

Answer: B — (b) Wrought iron

Chemical reason: Wrought iron is the purest of the listed commercial forms of iron, containing very little carbon compared with cast iron and steel.

357. Spelter is

- (a) Impure Cu
- (b) Impure Zn
- (c) ZnO
- (d) CuO

Answer: B — (b) Impure Zn

Chemical reason: The correct choice is Impure Zn. It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.





358. A copper coin is completely covered with a gold film and is placed in dilute HNO_3 . This will result in formation of

- (a) Gold nitrate
- (b) Copper nitrate
- (c) None of these
- (d) Purple of cassins

Answer: C — (c) None of these

Chemical reason: The correct choice is None of these. It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.

359. When zinc is added to CuSO_4 copper gets ppt. due to

- (a) Reduction of copper ions
- (b) Oxidation of copper ions
- (c) Hydrolysis of copper sulphate
- (d) Complex formation

Answer: A — (a) Reduction of copper ions

Chemical reason: Zinc is more reactive than copper, so Zn is oxidized to Zn^{2+} while Cu^{2+} is reduced to metallic Cu: $\text{Zn} + \text{Cu}^{2+} \rightarrow \text{Zn}^{2+} + \text{Cu}$. The copper precipitate is therefore formed by reduction of Cu^{2+} .

360. Addition of iron filings to CuSO_4 solution caused precipitation of Cu owing to the

- (a) Reduction of Cu^{+}
- (b) Oxidation of Cu^{+}
- (c) Reduction of Fe
- (d) Reduction of Fe^{++}

Answer: A — (a) Reduction of Cu^{+}

Chemical reason: This is determined by the standard composition or heat-treatment behaviour of the metal/alloy. The description in Reduction of Cu^{++} matches the relevant metallurgical process or alloy property.

361. Oxygen gas can be prepared from solid KMnO_4 by

- (a) Strongly heating the solid
- (b) Treating the solid with H_2 gas
- (c) Dissolving the solid in dil. H_2SO_4
- (d) Dissolving the solid in dil. HCl

Answer: A — (a) Strongly heating the solid

Chemical reason: The correct choice is Strongly heating the solid. It matches the established composition, oxidation state, and characteristic chemical behaviour of the substance named in the question; the other options are inconsistent with one or more of these features.

